Budget Guard: Calibrated Resource Control for Long-Horizon LLM Agents

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Abstract

Long-horizon LLM agents can consume substantial tokens, time, and money before recognising that a task is unlikely to succeed. Prior work on budget-aware agents shows the promise of early stopping, but agents' own remaining-budget estimates are poorly calibrated and cannot be trusted for control. We introduce Budget Guard, a model-agnostic, training-free wrapper that supplies two complementary controls at each trajectory prefix: conformal prediction intervals on remaining resource need, and a separately calibrated feasibility abort gate. The intervals answer how much may still be required; the gate answers whether continued execution is still worthwhile; they share features and data but are distinct controls with separate calibration targets and evaluation metrics. For interval calibration, Claude Opus 4.7 rollouts from Sokoban, Search-R1, and SWE-bench reach 0.90–0.96 empirical coverage offline (split-CQR) and 0.87–0.92 online (windowed-CQR) against a nominal 0.90, where agent self-reports reach only 0.51, 0.23, and 0.35; across 19 model–environment runs, windowed-CQR holds at least 0.85 coverage in 18 of 19, and in multi-resource Warehouse in-distribution Bonferroni joint calibration reaches 0.95 across time, capacity, and cost, though it falls to 0.71 under a model swap. For abort economics, in Warehouse only and without reading the remaining-budget intervals, thresholding the feasibility classifier at δ=0.10 avoids 86.3% of doomed-run spend with no false aborts observed in replay, and cuts Cost-of-Pass from \$1.61M to \$0.87M in simulated dollars per successful run at full pass retention; at the same δ, token-budget environments save far less (7.1% and 3.1% of doomed token spend on Sokoban and SWE-bench), and combining intervals into the abort rule is more conservative at fixed δ and does not increase doomed-run savings. All stopping results are counterfactual replays of logged rollouts, not live deployment. These results show that reliable budget awareness can be supplied as an external statistical control layer, without fine-tuning the underlying agent.

1 Introduction

LLM agents increasingly complete long-horizon tasks that require many reasoning steps, tool calls, and interactions with external environments [1], and those capabilities carry growing resource demands [2]. An agent may consume thousands of tokens, substantial wall-clock time, or real money before finishing a task [3], and it may spend much of that on redundant reasoning that does not improve the outcome [4–6]. When a trajectory eventually fails, most of that spending is wasted [3]. Controlling resource consumption without giving up task performance is therefore a practical obstacle to deploying agentic systems at scale [7–9].

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One difficulty is that an agent does not reliably know how much budget it still needs [3]. It can produce an estimate, but those estimates are uncalibrated, so they cannot be trusted for resource-control decisions.

This motivates Budget Guard, a model-agnostic framework for calibrated resource control in long-horizon LLM agents. Rather than teaching the underlying LLM budget awareness through fine-tuning, Budget Guard operates as an external statistical layer around an existing agent. At each trajectory prefix it uses observable information (resources consumed so far, progress, recent usage patterns, and the agent's own estimate when one is available) to support two decisions. No LLM is trained or fine-tuned; the only fitted components are lightweight gradient-boosted trees over usage-accounting features.

Rather than asking the agent for a single point estimate of remaining budget, Budget Guard produces an *interval* describing how much additional resource may still be required. Conformal prediction calibrates these intervals so that they achieve a desired empirical coverage rate under stated assumptions (exchangeability offline; rollout-safe online evaluation when labels arrive only at trajectory completion). Online recalibration absorbs temporal dependence and deployment shift when the underlying model or task type changes. Separately, a feasibility classifier, calibrated at the rollout level, supports an early-stopping rule with a user-chosen false-abort tolerance δ. These two components share features and data but serve different roles: the intervals answer "how much might remain?"; the gate answers "should we stop?" We evaluate interval calibration and abort economics independently, each against metrics matched to its role.

Our contributions are fourfold. (1) We introduce a conformal framework for calibrating progressive remaining-budget estimates that requires no LLM fine-tuning (§3, §5.1; Table 5). (2) We evaluate adaptive and sequential calibration under environment and model shift, where frozen calibration collapses from 0.92 to 0.10 coverage and online recalibration recovers much of it (§5.2; Figure 3). (3) We extend budget awareness to jointly constrained resources and censored trajectories (§3, §5.3, §5.5; Table 2). (4) We evaluate the feasibility abort gate independently of the remaining-budget intervals, reporting Warehouse abort performance together with token-environment results at the same δ and a classifier+interval variant that is more conservative and saves less at that operating point (§5.4; Figure 4, Table 3).

Budget Guard treats budget awareness as both a measurement problem and a control problem; the interval and abort signals are distinct and are calibrated and reported on separate metrics throughout.

2 Related work

BAGEN [13] formalises the estimation of an agent's remaining budget as budget awareness and evaluates whether an agent can estimate its remaining requirements mid-execution. A point estimate says nothing about its own uncertainty, and an interval is only useful if its coverage can be trusted; BAGEN takes intervals directly from the agent, and they are uncalibrated. Conformal prediction, by contrast, constructs intervals with specified coverage guarantees without distributional assumptions [14, 15]. BAGEN also asks the agent to judge, at different prefixes, whether successful completion is still feasible, and covers internal resources such as tokens alongside external ones such as time, money, and warehouse capacity across Sokoban, Search-R1, SWE-bench, and Warehouse.

The BAGEN results expose a gap between task capability and budget awareness: strong agents solve hard tasks while remaining poorly calibrated about what finishing will cost. BAGEN reports that early stopping on budget-awareness signals saves roughly 28–64% of tokens on failed trajectories, at a cost of roughly 1.6–4.2 percentage points of task success, with interval coverage reaching only 47% even after supervised fine-tuning and reinforcement learning [13]. The agent carries some signal about future requirements, but its raw estimates are not reliable enough for high-confidence control.

Doomed From the Start [16] monitors a model's internal signals to decide when a trajectory is doomed and terminates it early. In common with Budget Guard and unlike BAGEN, it applies a distribution-free calibrated abort gate with no fine-tuning, meeting recall targets of 90–97% on TextCraft and saving 47.1% (Qwen-2.5-7B) and 37.2% (Llama-3.2-3B) of inference compute at the 90% target. Because it reads internal states, it applies only to open-weight models.

A third line applies conformal prediction over an agent's answer. MiCP [17] allocates an error budget to each turn of a multi-turn reasoning process, letting an agent stop once it has gathered enough evidence while preserving an overall coverage guarantee. MiCP and Budget Guard share conformal machinery but calibrate different objects: MiCP calibrates coverage over the answer; Budget Guard calibrates coverage over the remaining resource budget. Budget Guard is also an external wrapper rather than a component of the reasoning loop, and additionally handles jointly constrained resources, censored trajectories, and a feasibility abort gate that we evaluate separately from the intervals (§5.4). We specify the architecture in §3 (Figure 1); Appendix A defines the background concepts used to interpret our results.

3 Method

Setup. At turn k of a rollout, Budget Guard observes only the prefix: per-turn resource usage so far and, when available, the agent's own estimate of its remaining budget. It outputs an interval for y, the resources the run still needs to finish successfully, at nominal miscoverage α=0.10. Two properties of this label shape the design. First, y is observable only once the rollout finishes, so no online method may consume a label mid-run. Second, y is defined only while the run can still finish: a doomed run censors its label rather than delaying it. The guard acts as a verifier: it need not trust the agent's self-report, because it wraps that self-report in a guarantee that holds regardless of its quality (Table 4).

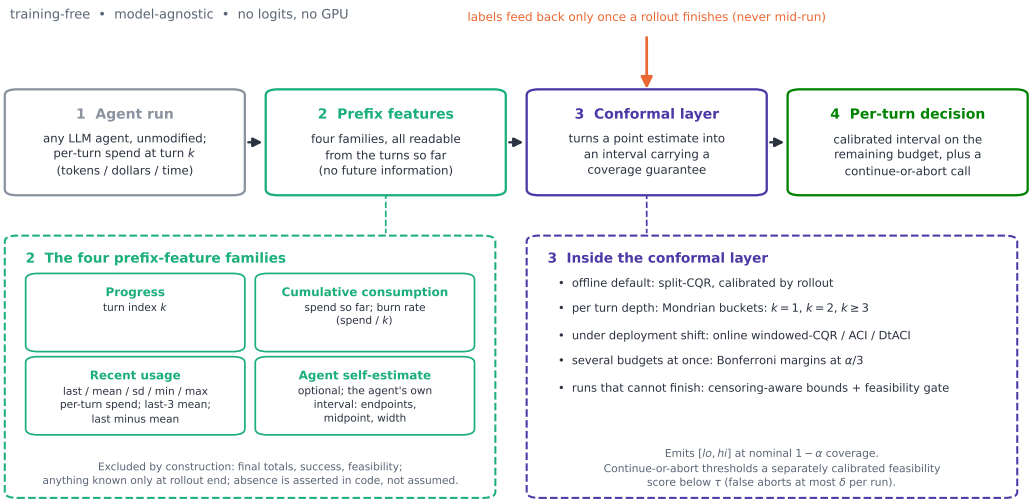


Figure 1: Budget Guard wraps an unmodified agent in an external calibration layer. Stages 1–4 share prefix features; stage 3 returns a calibrated remaining-budget interval (§§5.1–5.3), and a separately calibrated feasibility score drives continue-or-abort (§5.4). Warehouse abort economics use the feasibility gate alone and do not read the intervals. Labels become available only when a rollout finishes, so online recalibration never uses a still-running run.

Figure 1 sketches the two paths (interval calibration and feasibility abort), which share prefix features but are evaluated separately in what follows. The rest of this section specifies offline and online interval calibration for deployment shift, joint calibration when several resources bind at once, censoring-aware bounds behind a feasibility gate, and the economic stopping rule built on the gate alone.

Prefix features. We use four families of features, all computable at turn k with no access to the rest of the trajectory. Three come from the usage stream alone: progress (the turn index k); cumulative consumption (spend so far, and the burn rate spend/k); and recent-usage statistics (the last, mean, standard deviation, minimum and maximum per-turn spend, the mean of the last three turns, and the difference between the last turn and the mean). When available, a fourth family encodes the agent's self-estimate (interval endpoints, midpoint, and width). Features that depend on the rollout's

future (final totals, success, and feasibility) are excluded from the feature matrix and asserted absent in code. The guard needs no model weights, logits, or environment internals: every feature comes from usage accounting that provider APIs already report.

Calibrated intervals under deployment shift. The base predictor is a gradient-boosted quantile regressor [27]. We split calibration and test by rollout, never by prefix: prefixes of one rollout share its final total, so splitting them apart would leak each test rollout's answer into calibration. Frozen split-CQR holds only under exchangeability, which breaks under a model swap or a new task type. We therefore also recalibrate from the deployment stream's own finished rollouts three ways: ACI, which raises or lowers the working coverage level from recent misses [11]; DtACI, which removes ACI's hand-set step size [12]; and windowed-CQR, a conformal quantile over a sliding window of the 100 most recent scores, inspired by SPCI [18]; under non-exchangeability the coverage gap is governed by window-to-stream drift [19]. All observe labels rollout-safely: a rollout's labels enter the update state only after it finishes, and the state is frozen within a rollout. Revealing them immediately would use labels that do not yet exist at deployment and would inflate coverage, so we report only this deployment-faithful protocol. Two reporting rules follow. An unbounded upper endpoint covers everything and informs nothing, so we report the fraction of unbounded intervals beside every coverage number. And because a single global quantile is set by the many easy late-turn prefixes, a good average can hide under-coverage at the first turns, where an abort decision has the most leverage; we therefore also calibrate a separate conformal quantile per turn-depth bucket (Mondrian [20]): $k{=}1$, $k{=}2$, $k{\geq}3$.

Joint multi-resource calibration. When a run is constrained by several budgets at once (Warehouse: weeks, warehouse item-weeks, and dollars), the relevant event is joint: the run survives only if all three hold, and covering each at level α does not cover their conjunction, because the misses accumulate. Calibrating each resource at $\alpha/3$ (Bonferroni) restores a joint guarantee of $1{-}\alpha$ by the union bound, at the price of wider margins; this is our primary multi-resource calibration scheme. A tighter alternative shares one threshold by ranking the resources' errors jointly [21], but that rank is chosen on the same scores it then thresholds, so it carries no guarantee; we report it as an observation and rest the claim on Bonferroni. Under a model swap the joint calibration also moves: a rolling window of recent three-resource errors updates both the margins and the shared rank, again only once a run has finished.

Censored rollouts and a selective guarantee. When a run is doomed we never learn y, only that it exceeded what was left. We calibrate one-sided upper bounds five ways: treating censored scores as $+\infty$ (valid, but vacuous once the censored fraction exceeds α); substituting the known lower bound; dropping censored scores entirely (the naive comparison); reweighting by each run's censoring probability (IPCW [22, 23]); and charging a censored run only against the resource that stopped it. A feasibility gate, a classifier on the same prefix features, then issues intervals only on predicted-feasible prefixes, so coverage holds among the prefixes it admits rather than across all prefixes. This selective guarantee is what the construction supports; marginal coverage over all prefixes is reported as an empirical observation.

Economic stopping. The abort rule thresholds feasibility: stop at the first turn whose feasibility score falls below τ, the abort threshold. Because the rule aborts a rollout exactly when that rollout's minimum prefix score crosses τ, we calibrate at the rollout level, setting τ to the δ-quantile (with the finite-sample correction) of the minimum scores of feasible calibration rollouts. This bounds by δ (the false-abort tolerance) the probability of aborting a run that would have finished: a per-run guarantee matched to the sequential decision, not a per-prefix statement. Aborting a doomed run recovers its remaining spend, while aborting a feasible run forfeits its pass; both effects enter Cost-of-Pass [25], total spend per passing run, so the rule is evaluated in economic terms rather than as a coverage statistic. The rule is a calibrated binary gate in the sense of conformal risk control [26]: it thresholds the feasibility score and does not read the remaining-budget intervals. An interval-driven variant that aborts when spend-so-far plus the calibrated lower bound exceeds the budget is implemented in the released code; Warehouse economics below use the feasibility-gate rule only, and token environments also evaluate the combined variant (Table 3).

Table 1: The four evaluation environments; only Warehouse binds several resources at once, and only Warehouse censors labels. Token budgets are the BAGEN context budgets in tokens; Warehouse budgets are set per task. In Warehouse, half of the prefixes (640 of 1,280 per model) belong to runs that can no longer finish and drive the censored-calibration evaluation.

Environment	Task	Budgeted resource(s)	Budget	# Models
Sokoban	box-pushing puzzle	tokens	2,500 tokens	7
Search-R1	multi-turn retrieval QA	tokens	3,500 tokens	7
SWE-bench	repository-level bug fixing	tokens	6,772–21,711 (per run)	5
Warehouse	multi-week management sim.	time, capacity, cost	per task, joint	5

4 Experimental setup

Data and environments. All experiments use logged rollouts from BAGEN [13] at a pinned dataset revision; Table 1 summarises the four environments. Sokoban, Search-R1, and SWE-bench are single-resource token tasks. Sokoban and Search-R1 use uniform context budgets of 2,500 and 3,500 tokens; on SWE-bench the official budget varies by run (6,772, 7,060, 8,265, 10,472, and 21,711 tokens across the five models), and we read each run's value from its estimation record rather than applying a per-environment constant. The label is the number of tokens still needed to complete the task under BAGEN's turn-excluding-history accounting. Warehouse is a multi-week management simulation with three jointly binding budgets (time in weeks, capacity in item-weeks, and cost in simulated USD), and per-prefix ground truth for all three comes from the official estimation records. Half of its prefixes (640 of 1,280 per model) belong to runs that can no longer finish, which makes Warehouse our censoring test.

Protocol and baselines. Rollouts are split 50/25/25 into train, calibration, and test sets. We repeat every primary experiment with five random splits (seeds 0–4), assigning whole rollouts to a split rather than individual prefixes, so no rollout has its prefixes divided across sets; reported coverage and economics are means over the five seeds. Online methods consume the test stream in its natural order with labels revealed only at rollout boundaries. We evaluate three shift scenarios: cross-environment (calibrate on Sokoban; deploy on Search-R1 and SWE-bench), model swap (calibrate on Claude Opus 4.7; deploy on four other agents' SWE-bench streams), and an external-resource model swap on Warehouse. Seven comparators run on the same splits and seeds as the guard: the agent's own interval, a feature-free constant-quantile interval, split conformal on absolute residuals, a conformalized-agent interval, a guard-plus-self-estimate ablation, frozen split-CQR under shift, and hard caps for stopping. Coverage is reported against the 0.90 nominal level, always alongside the fraction of unbounded intervals. Appendix C gives the full protocol, baseline definitions, metric formulas, and implementation details.

Models. We evaluate seven agents (Claude Opus 4.7/4.6, Claude Sonnet 4.6, GPT-5.2-instant, GPT-5.2-Codex, Gemini 3.1 Pro, and Qwen3-235B). Crossed with environments this gives 19 token-budget model$\times$environment runs (7 Sokoban, 7 Search-R1, 5 SWE-bench) of 52–128 rollouts and 85–3,715 prefix samples each.

5 Results

We evaluate prediction intervals using empirical coverage, normalised width, and the fraction of unbounded intervals. Unbounded intervals raise coverage while carrying no information, so we hatch them in the figures and report the unbounded fraction beside every coverage number.

5.1 Within-run interval calibration

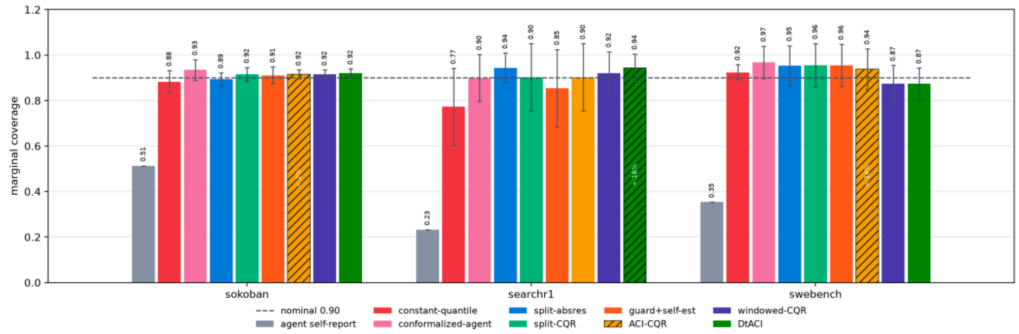


Figure 2: Conformal calibration closes almost all of the gap between agent self-reports and the 0.90 target, and windowed-CQR does so while maintaining bounded intervals throughout. Marginal coverage of the true remaining budget on Sokoban, Search-R1, and SWE-bench (Claude Opus 4.7 rollouts), α=0.10, averaged over five rollout-level split seeds; error bars are the standard deviation across those seeds. Hatched bars emit unbounded intervals on the annotated fraction of the stream, which raises coverage at the cost of informativeness. Agent self-report is a single value read from the logged records rather than a seed mean. Coverage, normalised width, unbounded fraction, and test-set sizes for all methods are in Table 5.

Figure 2 shows within-run remaining-budget calibration on Sokoban, Search-R1, and SWE-bench using Claude Opus 4.7 rollouts. Agent self-reported intervals are badly calibrated, covering only 0.51, 0.23, and 0.35 of prefixes against a 0.90 target. Offline conformal calibration largely closes that gap, with split-CQR reaching 0.92, 0.90, and 0.96. Online windowed-CQR reaches 0.92, 0.92, and 0.87 with no unbounded intervals in any of the three environments. ACI-CQR and DtACI sometimes reach higher coverage, but do so partly by emitting unbounded intervals on up to 16% of the stream, which shows that coverage alone is not a sufficient criterion. Across all 19 logged model–environment runs, windowed-CQR reaches at least 0.85 coverage in 18 of 19 and at least 0.90 in 11 of 19, against 14 and 8 for DtACI, on one split per run (Table 6). It is also sharper than the feature-free baseline on two of three environments: 22% narrower than constant-quantile on Sokoban and 55% narrower on SWE-bench, but 23% wider on Search-R1, where constant-quantile is the sharpest bounded method (Table 5). Adding the agent's own estimate to the guard's features does not help: guard+self-est is within 0.01 of split-CQR on Sokoban (0.91 against 0.92), identical on SWE-bench (0.96), and worse on Search-R1 (0.85 against 0.90).

5.2 Robustness under deployment shift

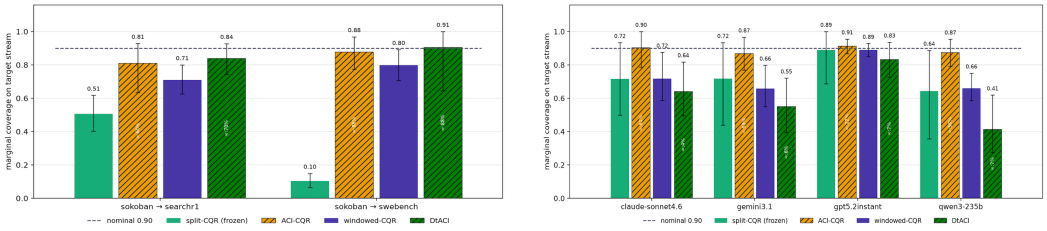


Figure 3: An environment change collapses frozen calibration and online recalibration recovers much of it; a model swap leaves every method below nominal coverage. Marginal coverage on the target stream. (a) A guard calibrated on Sokoban and deployed unchanged on Search-R1 and SWE-bench: frozen split-CQR falls to 0.51 and 0.10, and windowed-CQR recovers to 0.71 and 0.80. (b) A guard calibrated on Claude Opus 4.7 SWE-bench rollouts and deployed on four other agents: here frozen and online calibration are comparable on average (0.74 against 0.73) and neither reaches the target, though windowed-CQR varies far less across seeds. Hatched bars emit unbounded intervals on the annotated fraction of the stream (60–88% for the ACI-based methods in (a), and 22–61% (ACI-CQR) and 4–7% (DtACI) in (b)).

Figure 3 evaluates Budget Guard under deployment shift. Transferring a guard calibrated on Sokoban to Search-R1 and SWE-bench drops frozen split-CQR coverage to 0.51 and 0.10. Rollout-safe windowed-CQR partially recovers it, to 0.71 and 0.80, with no unbounded intervals; ACI and DtACI reach higher coverage but need unbounded intervals on 60–88% of predictions to get there. In the model-swap setting a guard calibrated on Claude Opus 4.7 is transferred to Sonnet, Gemini, GPT, and Qwen SWE-bench streams. Windowed-CQR reaches 0.66–0.89 there, and its seed-to-seed standard deviation is an order of magnitude smaller than frozen calibration's (0.006–0.011 against 0.122–0.224), so its failures are at least predictable. Online adaptation improves robustness under shift, but severe shift remains unsolved: no bounded-interval method recovers the nominal level, and high coverage has to be read together with interval informativeness.

5.3 Conditional and multi-resource calibration

Marginal coverage can mask under-coverage in conditional slices. On Sokoban, split-CQR reaches 0.92 marginal coverage but only 0.86 at the first turn, exactly where an abort decision has the most leverage; Mondrian calibration by turn depth raises first-turn coverage to 0.93. On Search-R1, Mondrian is neutral to slightly negative relative to split-CQR; on SWE-bench the marginal-to-first-turn gap is already small and Mondrian is neutral to slightly positive (Table 7), yielding conditional validity rather than a uniform improvement.

Table 2: Independent margins under-cover the joint event, Bonferroni fixes it in distribution, and nothing with a guarantee survives a model swap. Warehouse coverage against a nominal 0.90, mean over five rollout-level split seeds; above 0.90, coverage is bought with wider margins. Per-resource columns and the in-distribution joint column are Claude Opus 4.7; the final column is the joint event averaged over four swap targets. Bold marks the highest joint coverage in each column. "–" marks a scheme not run in that setting.

	per-resource			joint	
Calibration scheme	time	capacity	cost	in-dist.	model swap
agent self-report	0.87	0.67	0.71	0.42	0.34
independent margins (α)	0.99	0.89	0.92	0.80	–
Bonferroni ($\alpha/3$)	1.00	0.98	0.97	**0.95**	0.71
empirical copula	0.99	0.94	0.96	0.89	0.62
adaptive copula (online)	–	–	–	–	**0.87**

5.4 Economic impact

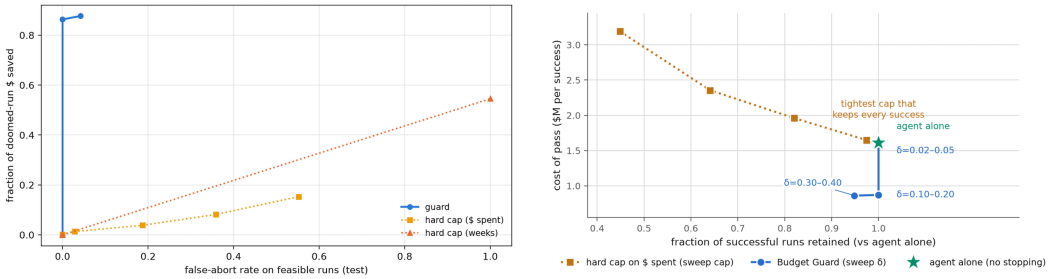


Figure 4: Warehouse economic impact from the feasibility gate alone, without remaining-budget intervals. At δ=0.10 the classifier-only rule avoids 86.3% of doomed-run spend with no observed false aborts, which no hard cap achieves at any setting. Simulated USD, mean over five seeds. (a) Every hard cap that saves more than 0.2% of doomed spend also stops runs that would have succeeded; the gate occupies the favourable region of the savings–false-abort trade-off. (b) Cost-of-Pass against the fraction of successful runs retained: the gate roughly halves cost per pass while retaining every observed success, whereas no cap improves on the unstopped baseline. Both panels are counterfactual replays of logged rollouts, not live deployments.

Table 2 reports the multi-resource case. In Warehouse, agent self-reports reach 0.42 joint coverage across time, capacity, and cost. Calibrating each resource independently at α lifts the joint event only to 0.80, reflecting that independent margins do not compose to joint coverage. Bonferroni at $\alpha/3$ restores 0.95, and the empirical copula reaches 0.89 with narrower margins but no guarantee. Under an external-resource model swap, however, frozen Bonferroni falls to 0.71, below its nominal level, and below the 0.87 reached by the un-guaranteed adaptive copula, which recalibrates from finished rollouts. We report both the guaranteed and empirical methods.

Figure 4 evaluates early stopping in Warehouse using the classifier-only feasibility gate. At $\delta{=}0.10$ the calibrated gate avoids 86.3% of doomed-run spend with 0% observed false aborts. Hard spending caps achieve almost nothing at zero false aborts (the tightest cap that preserves every successful run saves 0.2% of doomed spend), and dollar caps reach meaningful savings only at false-abort rates of 19–55%; the only week cap that saves anything (54.5%) aborts every feasible run. In the Cost-of-Pass analysis the guard reduces estimated cost per successful trajectory from \$1.61M to \$0.87M, a reduction of 46%, while retaining all observed successful runs. No hard cap improves Cost-of-Pass relative to the unstopped baseline by more than a rounding error: discarding passes raises cost per pass, to \$3.19M at the tightest cap.

Table 3: Abort performance at $\delta{=}0.10$ (mean over five seeds). Warehouse uses the classifier-only gate (native USD). Token abort is evaluated on Sokoban and SWE-bench only; both also evaluate a classifier+interval variant. Combining intervals reduces doomed savings at this operating point.

Environment	Rule	False abort	Doomed spend saved	Uses intervals?
Warehouse	feasibility only	0.0%	86.3%	No
Sokoban	feasibility only	5.8%	7.1%	No
Sokoban	feasibility+interval	5.8%	1.9%	Yes
SWE-bench	feasibility only	0.0%	3.1%	No
SWE-bench	feasibility+interval	0.0%	0.0%	Yes

Table 3 reports abort performance across environments and rules. At the same $\delta{=}0.10$, Sokoban and SWE-bench save only 7.1% and 3.1% of doomed token spend under the classifier-only rule, with false-abort rates of 5.8% and 0%. Combining the interval trigger makes the policy more conservative at fixed δ (Sokoban savings fall to 1.9%; SWE-bench to 0%) and does not recover Warehouse-scale savings. The 86.3% figure is therefore a Warehouse result for the feasibility gate alone, not a general Budget Guard savings rate, and it does not use conformal remaining-budget intervals. Two further cautions: the dollars are simulated Warehouse dollars, not comparable to BAGEN's 28–64% token savings; and the observed 0% false-abort rate should be read as consistent with the $\delta{=}0.10$ bound on a small test set, not as proof that the true rate is zero.

5.5 Censored calibration

Roughly half of Warehouse prefixes are censored: the run can no longer finish, so its true completion requirement is never observed. Dropping those observations yields only 0.50 coverage on the censored prefixes where cost is the binding resource, so failed trajectories cannot be treated as missing data. Treating censored scores as $+\infty$ is valid but vacuous at this censoring rate, and the informative treatments trade validity for sharpness in ways the appendix makes explicit. Feasibility-gated selective intervals offer the other lever: abstaining on predicted-doomed prefixes holds coverage at 0.92–0.95 for capacity and cost, and at 1.00 for time, on the prefixes it keeps, at the cost of keeping between 51% and 89% of them (Table 8). The gate is swept over its own prefix-level tolerance δ_{gate}, which is distinct from the rollout-level δ of §5.4. One caveat governs how these numbers should be read: the recovered labels used to score censored prefixes are exact only for time, and inherit a week-proportional interpolation with 12–27% median relative error for capacity and cost, so those two are coverage against an approximate label rather than ground truth.

6 Discussion and limitations

Our joint results rest entirely on Warehouse. Bonferroni calibration reaches 0.95 joint coverage there, but a single multi-resource environment cannot support a claim that the approach generalises, and

§5.3 shows that guarantee is not robust to a model swap. The 86.3% doomed-spend figure likewise does not transfer to token environments at δ=0.10 (Table 3).

Our breadth figures are single-split counts: on seed 0, windowed-CQR reaches at least 0.85 coverage in 18 of 19 runs, not all 19. Search-R1 leaves about 24 test prefixes per seed, which is why its standard deviations reach 0.10–0.17. Coverage alone does not separate a useful interval from a trivially wide one; we therefore report width and unbounded fraction throughout. Censored capacity/cost results are measured against an interpolated label. Finally, savings and Cost-of-Pass are counterfactual replays: they assume adding the guard would not change agent behaviour and that the sample is representative.

Broader impact. The intended effect is to reduce wasted compute, energy, and money on agent runs that cannot succeed; the Warehouse feasibility gate at δ=0.10 illustrates that potential in one environment (86.3% of doomed-run spend avoided in replay, without using remaining-budget intervals). The corresponding risk is that a stopping threshold set too aggressively terminates runs that would have finished; δ calibration exists to bound that risk at a user-chosen level, and the counterfactual caveat above means a deployer should not treat the reported savings as a guaranteed live return when choosing δ, nor assume token environments will see Warehouse-scale savings.

7 Conclusion

Budget Guard shows that reliable budget awareness can be supplied as an external control layer for long-horizon LLM agents, with no fine-tuning of the underlying model. Conformal calibration of progressive remaining-budget estimates yields 0.90–0.96 empirical coverage offline (split-CQR) and 0.87–0.92 online (windowed-CQR) against a 0.90 target on Claude Opus 4.7 rollouts, and holds at least 0.85 coverage in 18 of 19 model$\times$environment runs. Joint multi-resource calibration works in distribution but does not survive a model swap. Separately, a calibrated feasibility gate (not the remaining-budget intervals) avoids 86.3% of doomed-run spend in Warehouse replay at δ=0.10, while token-budget environments yield much smaller savings at the same operating point. Budget awareness can therefore serve not only as a measurement but as a control signal during execution.

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A Background and guarantees

This appendix defines the concepts needed to interpret the results.

- **Coverage.** The fraction of cases in which the true remaining budget falls inside the predicted interval. Coverage of 0.90 means the true value is contained in about 90% of cases.
- **Calibration set.** A held-out set of completed trajectories used to set interval widths and stopping thresholds.
- **Rollout (run).** One complete agent attempt at a task.
- **Turn (k).** One agent step within a rollout.
- **Prefix.** The partial trajectory containing all turns completed through turn k.
- **$\hat{q}$.** The conformal threshold: a learned error margin added to the model's predictions to expand them into an interval. Throughout, $\hat{q}$ is the conformal threshold and $\hat{f}_{\mathrm{lo}}$, $\hat{f}_{\mathrm{hi}}$ are the quantile regressor's own predictions.

Table 4: What each component guarantees, and what it does not. Statements we do not have are listed as such, so a reader can tell a theorem from an observation.

Method	Role in Budget Guard	Guarantee and scope
split-CQR	CQR intervals conformalized on a by-rollout calibration split; our offline default [10].	Marginal coverage $\geq 1-\alpha$ over a random (rollout, prefix) pair, under exchangeable rollouts. Pooling a rollout's prefixes is an empirical evaluation, not the guaranteed object [15].
Mondrian-CQR	split-CQR calibrated separately per turn-depth bucket [20].	Coverage $\geq 1-\alpha$ within each bucket, under exchangeability within the bucket; thin buckets fall back to the global quantile.
ACI / DtACI	Adaptive coverage levels recalibrated from the stream's finished rollouts [11, 12].	Long-run average coverage approaches $1-\alpha$. May emit unbounded intervals; we clamp the working level at 0.999 rather than emit empty sets, a conservative deviation.
windowed-CQR	Conformal quantile over a sliding window of the 100 most recent scores; an SPCI-inspired simplification [18].	No finite-sample guarantee; coverage gap bounded by window-to-stream drift [19]. Evaluated empirically.
Bonferroni joint	Calibrates each resource at $\alpha/3$ so all hold at once; our primary multi-resource calibration scheme.	Joint coverage $\geq 1-\alpha$ by the union bound; no data-driven selection; conservative, so realised coverage exceeds the target in distribution.
empirical copula	Shared copula rank across resources [21]; an adaptive variant updates the rank per finished rollout.	Empirical only: the rank is selected on the same scores it thresholds. Reported alongside Bonferroni, not as the guaranteed claim.
censoring-aware upper bounds	One-sided bounds for doomed runs [22–24].	One-sided upper coverage on the censored subset. $+\infty$ always valid but vacuous once the censored fraction exceeds α; the others empirical or assumption-dependent.
feasibility-gated intervals	A classifier on the same prefix features issues intervals only on predicted-feasible prefixes.	Coverage $\geq 1-\alpha$ conditional on passing the gate: selective, not marginal. Marginal coverage reported empirically.
stopping rule (τ)	Abort at the first turn whose feasibility score drops below τ [26]; scored by Cost-of-Pass [25].	False-abort probability $\leq \delta$ per run, from rollout-level calibration on feasible runs.

- α**.** The allowed miscoverage rate; the target coverage is $1-\alpha$. We use α=0.10 throughout. A smaller α raises the target and generally widens intervals.
- **Width.** The size of the predicted range. Narrower is more informative at equal coverage: $[100, 115]$ is more useful than $[100, 300]$. An interval extending to infinity attains coverage while saying nothing.
- δ**.** The false-abort tolerance for the stopping rule. A larger δ permits greater savings on doomed runs while accepting more risk of aborting successful ones.
- τ**.** The abort threshold. The guard aborts at the first turn whose feasibility score falls below τ. It is not chosen by hand: it is read off the calibration rollouts as the value holding the false-abort rate at or below δ, so a smaller δ produces a more cautious τ.
- **Offline.** Calibrates once on completed runs and stays fixed during deployment.
- **Online.** Updates calibration after each completed run, using its newly observed remaining-budget labels.

Table 4 states, component by component, what is guaranteed and under what assumption. Four rows carry no finite-sample guarantee at all (windowed-CQR, the empirical copula, the censoring-aware bounds other than $+\infty$, and the marginal rather than selective reading of the gated intervals), and every claim resting on those is reported in this paper as an empirical observation.

B Per-run breadth results

Table 6 lists every model×environment run behind the breadth claims in §5.1; the "18 of 19" and "11 of 19" counts can be read straight off the windowed-CQR column. Table 7 gives conditional coverage by turn depth, Table 8 the feasibility-gate sweep, and Table 9 the censored upper-bound

Table 5: Full within-run results: every conformal method lands within 0.05 of the 0.90 target except the feature-free constant-quantile baseline on Search-R1, where the agent's own intervals miss by 0.39–0.67; windowed-CQR gives the narrowest bounded intervals on Sokoban and the second-narrowest on SWE-bench, while constant-quantile is sharpest on Search-R1. Claude Opus 4.7, nominal 0.90, mean$\pm$ standard deviation over five rollout-level split seeds. Normalised width is interval width divided by the central 80% range of the labels (lower is sharper). "Unb." is the percentage of predictions with an infinite upper endpoint; width is shown as "–" where at least one seed produced a majority of unbounded intervals. n_{test} is test prefixes per seed.

Env	Method	Coverage	Norm. width	Unb. (%)	n_{test}
Sokoban	agent self-report	0.51	–	–	–
	constant-quantile	0.88 ± 0.05	1.10	0	87
	split-absres	0.89 ± 0.03	1.12	0	87
	split-CQR	0.92 ± 0.03	0.88	0	87
	conformalized-agent	0.93 ± 0.05	1.44	0	87
	guard+self-est	0.91 ± 0.04	0.91	0	87
	ACI-CQR	0.92 ± 0.02	0.88	2	87
	windowed-CQR	0.92 ± 0.02	0.86	0	87
	DtACI	0.92 ± 0.02	0.88	0	87
Search-R1	agent self-report	0.23	–	–	–
	constant-quantile	0.77 ± 0.17	0.90	0	24
	split-absres	0.94 ± 0.06	1.60	0	24
	split-CQR	0.90 ± 0.15	1.14	0	24
	conformalized-agent	0.90 ± 0.10	1.96	0	24
	guard+self-est	0.85 ± 0.17	1.07	0	24
	ACI-CQR	0.90 ± 0.15	1.14	0	24
	windowed-CQR	0.92 ± 0.10	1.10	0	24
	DtACI	0.94 ± 0.06	–	16	24
SWE-bench	agent self-report	0.35	–	–	–
	constant-quantile	0.92 ± 0.03	2.87	0	161
	split-absres	0.95 ± 0.09	2.56	0	161
	split-CQR	0.96 ± 0.10	2.11	0	161
	conformalized-agent	0.97 ± 0.07	2.23	0	161
	guard+self-est	0.96 ± 0.09	1.97	0	161
	ACI-CQR	0.94 ± 0.09	–	14	161
	windowed-CQR	0.87 ± 0.08	1.29	0	161
	DtACI	0.87 ± 0.07	1.22	0	161

calibrations. Every number is read from the generated result tables, which one command regenerates and two checkers verify.

C Protocol, baselines, metrics, and implementation

This appendix records what §4 summarises: how the data is split, what each baseline is, how each reported number is computed, and the settings under which all of it runs. Every value below is fixed in the released code, and the parenthetical names are the modules that implement it.

C.1 Splits and seeds

Every split is by rollout, never by prefix (`split.group_split`). For a given seed the unique rollout indices are permuted with `numpy.random.default_rng(seed)` and allocated 50/25/25 to train, calibration and test; every prefix follows its rollout. Prefixes of one rollout share that rollout's final total, so splitting them apart would place each test rollout's answer into calibration. A split that is requested but comes out empty raises an error rather than degrading quietly: with an empty calibration set the conformal quantile of zero scores is $+\infty$, every interval becomes $[0, \infty)$, and coverage reads a meaningless 1.00 with no other symptom. Reported numbers are means over five seeds, 0 through 4.

Table 6: Breadth across all 19 token-budget runs: windowed-CQR clears 0.85 on 18 of them and 0.90 on 11, while the agents' own intervals clear neither anywhere. Marginal coverage against nominal 0.90 on a single rollout-level split (seed 0). Four decimals are shown on the three guard columns so the two counts above are directly checkable; at three decimals SWE-bench/Gemini DtACI rounds to 0.850 and appears to clear 0.85 when it does not. "–" marks a run whose records carry no agent interval. †Warehouse is listed for dataset composition only; its 0.424 is joint coverage across three resources (Table 2), not a token-budget marginal.

Environment	Model	Rollouts	Prefixes	Agent	split-CQR	windowed-CQR	DtACI
	Claude Opus 4.6	116	410	0.446	0.8077	0.8750	0.8846
	Claude Opus 4.7	123	355	0.513	0.8667	0.9000	0.9000
	Claude Sonnet 4.6	126	400	0.524	0.9623	0.9151	0.9528
Sokoban	Gemini 3.1 Pro	126	389	0.125	0.9425	0.9310	0.9310
	GPT-5.2-instant	128	661	0.481	0.8176	0.8986	0.8716
	GPT-5.2-Codex	128	592	–	0.9933	0.9600	0.9467
	Qwen3-235B	126	1,038	0.090	0.9268	0.9024	0.8862
	Claude Opus 4.7	66	87	0.233	1.0000	1.0000	1.0000
	Claude Sonnet 4.6	64	102	0.386	1.0000	0.9200	1.0000
	Claude Opus 4.6	52	90	0.478	0.9130	0.9130	0.9130
Search-R1	GPT-5.2-instant	125	324	0.198	0.8256	0.8605	0.8488
	GPT-5.2-Codex	126	262	0.202	0.9649	0.9298	0.8772
	Gemini 3.1 Pro	83	85	0.202	0.8182	0.8636	0.8182
	Qwen3-235B	124	1,079	0.025	0.9259	0.9259	0.8765
	Claude Opus 4.7	64	712	0.354	0.9926	0.9185	0.9037
	Claude Sonnet 4.6	64	1,258	0.251	0.9101	0.8705	0.8957
SWE-bench	Gemini 3.1 Pro	64	2,327	0.254	0.8497	0.8899	0.8497
	GPT-5.2-instant	64	1,313	0.385	0.8840	0.8589	0.8370
	Qwen3-235B	60	3,715	0.034	1.0000	0.7727	0.8271
Warehouse	Opus 4.7 + 4 swap targets	64 ea.	1,280 ea.	0.424†	–	–	–

Table 7: Marginal coverage hides a first-turn deficit on Sokoban, and Mondrian calibration is what removes it. Coverage by turn-depth bucket, nominal 0.90, mean over five seeds. Search-R1 rollouts are short enough that no $k \geq 3$ bucket exists.

Environment	Method	Marginal	$k=1$	$k=2$	$k \geq 3$
Sokoban	split-CQR	0.92	0.86	0.93	0.96
Sokoban	Mondrian-CQR	0.94	0.93	0.96	0.94
Search-R1	split-CQR	0.90	0.91	0.89	–
Search-R1	Mondrian-CQR	0.89	0.89	0.89	–
SWE-bench	split-CQR	0.96	0.99	0.99	0.95
SWE-bench	Mondrian-CQR	0.97	0.99	0.99	0.96

C.2 Rollout-safe online evaluation

Online methods consume the test rows in their natural trajectory order, rollout by rollout and turn by turn. A prefix's label is the resource the run still needs to finish, which cannot be observed until the rollout ends, so a rollout's labels enter the update state only at its boundary and the state is frozen within a rollout (`conformal.aci_cqr`, `windowed_cqr`, `dtaci`, each taking a `groups` argument). The code rejects test streams whose rollout ids are not contiguous blocks: an id that reappears after a boundary would silently absorb a still-running rollout's labels and inflate coverage with no visible symptom. Revealing labels per prefix instead would use information a deployed guard does not have, so we report only the rollout-safe protocol.

C.3 Base predictor and features

The point and quantile predictors are LightGBM models [27] with `n_estimators`$=200$, `learning_rate`$=0.05$, `num_leaves`$=15$ and `min_child_samples`$=10$. CQR fits two quantile regressors, at $\alpha/2$ and $1-\alpha/2$; the absolute-residual baseline and the interval-driven stopping variant use a squared-error regressor; the feasibility gate is an LGBM classifier with the same settings. No hyperparameter search was run: these are the defaults fixed at the start of the project and used unchanged for every experiment, baseline included.

Table 8: The feasibility gate trades retained volume for coverage on what it keeps: abstaining on predicted-doomed prefixes holds 0.92–0.95 coverage on capacity, where marginal calibration reaches only 0.89. Warehouse, mean over five seeds. "Kept" is the fraction of prefixes the gate admits, "caught" the fraction of doomed prefixes it correctly abstains on, "false abort" the fraction of feasible prefixes it wrongly rejects, and "coverage" the interval coverage among admitted feasible prefixes. Time binds in only 30% of censored runs and the gate catches almost all of them, so its row is flat. δ here is the gate's prefix-level tolerance, not the rollout-level false-abort tolerance of §5.4.

Resource	δ	Kept (↑)	Caught (↑)	False abort (↓)	Coverage (↑)
time (weeks)	0.02	0.88	0.93	0.00	1.00
	0.05	0.88	0.93	0.00	1.00
	0.10	0.88	0.93	0.00	1.00
	0.20	0.88	0.93	0.00	1.00
	0.30	0.88	0.93	0.00	1.00
capacity (item-weeks)	0.02	0.89	0.31	0.04	0.95
	0.05	0.78	0.64	0.05	0.95
	0.10	0.70	0.74	0.07	0.95
	0.20	0.57	0.82	0.19	0.95
	0.30	0.51	0.85	0.23	0.94
cost (USD)	0.02	0.86	0.60	0.05	0.92
	0.05	0.81	0.69	0.10	0.92
	0.10	0.80	0.72	0.10	0.92
	0.20	0.74	0.75	0.21	0.93
	0.30	0.74	0.76	0.21	0.93

Table 9: Dropping censored runs under-covers exactly the expensive failures, and every informative repair trades validity for sharpness. One-sided upper-bound coverage on Warehouse at nominal 0.90, split by whether the resource was the binding constraint. $+\infty$ is always valid and always vacuous; "drop" is the naive comparison. Kaplan–Meier is a diagnostic, not a calibration option. Coverage for capacity and cost is measured against an interpolated label (12–27% median relative error), not ground truth.

Resource	Treatment	Feasible (↑0.90)	Cens. binding	Cens. non-binding	Binding frac.
time (weeks)	drop (naive)	1.00	1.00	1.00	0.30
	$+\infty$	1.00	1.00	1.00	0.30
	substituted bound	1.00	1.00	1.00	0.30
	IPCW	0.96	0.96	0.96	0.30
	per-resource bound	0.90	0.90	0.90	0.30
	Kaplan–Meier (diagnostic)	1.00	1.00	1.00	0.30
capacity (item-weeks)	drop (naive)	0.94	0.77	0.98	0.34
	$+\infty$	1.00	1.00	1.00	0.34
	substituted bound	0.99	0.98	1.00	0.34
	IPCW	0.93	0.75	0.98	0.34
	per-resource bound	0.91	0.74	0.98	0.34
	Kaplan–Meier (diagnostic)	0.94	0.78	0.99	0.34
cost (USD)	drop (naive)	0.88	0.50	0.97	0.42
	$+\infty$	1.00	1.00	1.00	0.42
	substituted bound	1.00	0.99	1.00	0.42
	IPCW	0.88	0.50	0.96	0.42
	per-resource bound	0.86	0.46	0.94	0.42
	Kaplan–Meier (diagnostic)	0.92	0.58	0.98	0.42

The feature matrix (`features.build_xy`) is: the turn index k; cumulative spend; burn rate (spend/k); the last, mean, standard deviation, minimum and maximum per-turn spend; the mean of the last three turns; and the difference between the last turn and the mean. When the agent emits its own interval, four columns are added (the two endpoints, the midpoint and the width), and a missing estimate is left as `NaN`, which LightGBM handles natively. Quantities known only at rollout end (final totals, success, feasibility) are listed in a leak set and asserted absent from the matrix at build time, so the exclusion is enforced rather than intended.

C.4 Conformal methods and their settings

Every method uses the finite-sample corrected quantile: with n calibration scores the level is $\lceil (n+1)(1-\alpha) \rceil / n$, read with higher interpolation, returning $+\infty$ when that level exceeds 1. All intervals are clipped below at zero, since a remaining budget cannot be negative. α=0.10 throughout.

split-CQR uses the score $\max(\hat{f}_{\mathrm{lo}} - y,\ y - \hat{f}_{\mathrm{hi}})$ and one global $\hat{q}$; it is the offline default. Mondrian-CQR applies the same score with one $\hat{q}$ per turn-depth bucket, split at k=1, k=2 and $k \geq 3$; a bucket holding fewer than 10 calibration scores falls back to the global $\hat{q}$.

ACI updates the working level by $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$. γ is selected on calibration data only: the calibration rollouts are split in half, ACI is streamed over the second half under the same rollout-safe revelation used at test time, and the $\gamma \in \{0.005, 0.01, 0.02, 0.05\}$ whose coverage lands closest to $1-\alpha$ is kept. One deliberate deviation from canonical ACI: where the canonical algorithm emits the empty set once $\alpha_t \geq 1$, we clamp the quantile lookup at 0.999 and emit a finite interval. The deviation is conservative in both coverage and the error feedback, and we state it rather than diverge silently.

windowed-CQR keeps a rolling window of the 100 most recent conformity scores, warm-started from calibration, and reads the interval half-width off that window's empirical quantile, so the score distribution itself tracks drift rather than reweighting a frozen calibration set. SPCI [18] is the inspiration, not the implemented method.

DtACI aggregates ACI experts at $\gamma \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$ with learning rate $\eta = 2.0$ and uniform mixing $\sigma = 0.01$. After each revealed label, experts are scored by pinball loss against β_t, the largest miscoverage level whose quantile would still have covered the observed score, and the working level is the weight-averaged α_t.

C.5 Baselines

Seven comparators run on the same splits and seeds as the guard.

1. **Agent self-report.** BAGEN's own per-sample indicators of whether the agent's interval contained the truth, read from the official estimation records. No calibration is applied.
2. **Constant-quantile.** Feature-free: the empirical $[\alpha/2, 1-\alpha/2]$ interval of the calibration labels, identical at every test prefix.
3. **Split-absres.** A LightGBM point prediction plus or minus the calibrated absolute-residual quantile.
4. **Conformalized-agent.** The agent's own interval midpoint used as the point predictor and conformalized on $|y - \mathrm{mid}|$.
5. **Guard+self-est.** split-CQR on the guard features augmented with the agent's estimate columns.
6. **Frozen split-CQR.** split-CQR calibrated on the source environment or model and applied unchanged to the target stream.
7. **Hard caps.** Abort once spend so far exceeds a fixed cap. Dollar caps are swept at the 20th, 35th, 50th, 65th and 80th percentiles of realised run totals, and week caps at $0.5\times$ the 25th, $0.75\times$ the 50th and $1.0\times$ the 75th percentile of the task's time budget.

Every baseline sees the same training split, the same calibration split, the same test stream and the same seeds as the guard. BAGEN's SFT+RL estimator is not included as a baseline: it is a fine-tuned agent rather than an external wrapper, and re-running it would require training the underlying LLM, which is exactly what this work avoids.

C.6 Metrics

Coverage is the fraction of test prefixes whose interval contains the true remaining requirement. Normalised width is the interval width divided by the central 80% range of the labels, $q_{0.9} - q_{0.1}$, which makes environments whose token scales differ by orders of magnitude comparable; the median width is used wherever a method can emit unbounded intervals, and the across-seed summary

is reported as "–" where at least one seed's median is itself infinite. Unbounded fraction is the share of predictions with an infinite upper endpoint, and it appears beside every coverage number.

Confidence intervals come from a cluster bootstrap that resamples whole rollouts rather than prefixes, with $B = 500$ resamples; per-sample binomial intervals are too tight here, because the prefixes of one rollout are correlated. Across seeds, the per-seed bootstrap draws are pooled and the 5th and 95th percentiles of the pooled distribution are reported as a 90% interval; these are recorded in the released result tables rather than plotted. The $\pm$ shown in Table 5 and the error bars in Figure 2 are both the standard deviation across the five seeds.

For the stopping rule, the false-abort rate is the fraction of feasible test rollouts that were stopped, and the doomed-savings fraction is the spend avoided on rollouts that could not finish divided by the spend those rollouts would otherwise have incurred. Cost-of-Pass is total spend under a policy divided by the number of runs that still pass [25]. All economic results in this paper are Warehouse, which is priced natively in simulated dollars; token-environment economics appear only in Table 3.

C.7 Multi-resource and censored calibration

Warehouse binds time in weeks, capacity in item-weeks and cost in simulated dollars at once. Independent calibration sets each margin at α; Bonferroni sets each at $\alpha/3$, which restores a joint guarantee of $1-\alpha$ by the union bound and is the primary multi-resource calibration scheme. The empirical-copula variant shares one rank across resources and is tighter, but that rank is selected on the same calibration scores it then thresholds, so it carries no guarantee and is reported as an observation only. Under a model swap, an adaptive variant recomputes both the margins and the shared rank from a rolling window of recent three-resource error vectors, updated only once a rollout has finished.

For censored runs the one-sided upper bound is calibrated five ways (`censored.upper_qhat`): drop; infinity; bound; ipcw; and resource_bound. A Kaplan–Meier estimator [24] is a diagnostic rather than a sixth option: it returns $+\infty$ exactly when the upper tail is entirely censored. Table 9 reports the five calibrations together with the Kaplan–Meier diagnostic. Censored-stratum coverage for capacity and cost is coverage against an approximate interpolated label (12–27% median relative error), not against ground truth.

The feasibility gate is an LGBM classifier on the same prefix features. It issues intervals only on predicted-feasible prefixes, so the resulting guarantee is selective. The gate is swept at its own prefix-level tolerance $\delta_{\text{gate}} \in \{0.02, 0.05, 0.10, 0.20, 0.30\}$, which is a different quantity from the rollout-level false-abort tolerance δ of §5.4; Table 8 reports the full sweep.

C.8 Stopping rule and economics

The reported rule aborts at the first turn whose feasibility score falls below τ. Because the rule aborts a rollout exactly when that rollout's minimum prefix score crosses τ, τ is calibrated at the rollout level: it is set to the score at rank $\lfloor (n+1)\delta \rfloor$ among the sorted minimum scores of the n feasible calibration rollouts, which bounds by δ the probability of aborting a run that would have finished. δ is swept over $\{0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$ for the frontier; δ=0.10 is the operating point quoted in §5.4 and the abstract.

An interval-driven variant, which aborts when spend so far plus the calibrated lower bound exceeds the budget, is implemented in the released code and calibrated the same way at rollout level. It is not the rule behind the reported Warehouse economics, and we do not claim the two are equivalent. The economic numbers are counterfactual replays, not live measurements: each completed run is truncated where the guard would have aborted it and the avoided spend is summed.

C.9 Data, environment, and reproduction

All experiments use logged rollouts from BAGEN [13], fetched from its public Hugging Face release at the pinned revision `58189576e54b675fdd0e1d6c1c9f189c2992732f`. The extractor reproduces BAGEN's turn-excluding-history token accounting exactly, with 0 of 355 mismatches against the official per-sample ground truth on the reference run.

The whole pipeline runs inside a container whose base image is pinned by digest (`python:3.12-slim-bookworm`), with numpy 2.5.1, pandas 3.0.3, LightGBM 4.6.0, scikit-learn 1.9.0, scipy 1.18.0 and matplotlib 3.11.0 resolved from a hash-locked requirements file. The guard is CPU-only: no GPU, no model weights, and no access to logits or environment internals. No LLM is trained, fine-tuned or queried by us at any stage. A single command regenerates every figure and table in the repository and then machine-verifies the paper against what it just produced, exiting non-zero if any reported claim no longer matches.

NeurIPS Paper Checklist

1. **Claims. [Yes]** Every reported number in the abstract and Section 1 is produced by an experiment in this paper. The abstract distinguishes interval calibration from abort economics, states that the 86.3% savings and Cost-of-Pass figures are Warehouse-only and do not use remaining-budget intervals, and reports token-environment abort results at the same δ. Scope limits are stated throughout: Claude Opus 4.7 for within-run coverage, 0.95 joint coverage in-distribution only (0.71 under model swap), and counterfactual replay rather than live deployment.
2. **Limitations. [Yes]** Section 6 states that the joint multi-resource result rests entirely on one environment and does not survive a model swap, that the breadth counts come from a single split rather than a seed average, that marginal coverage can hide under-coverage in slices such as early turns, and that the economic figures are counterfactual replays rather than live measurements. Appendix C adds the assumption-level caveats and Table 4 marks which components carry no finite-sample guarantee.
3. **Theory assumptions and proofs. [N/A]** The paper states and proves no new theoretical results. Every guarantee is imported from prior work and Table 4 records what is guaranteed for each component.
4. **Experimental result reproducibility. [Yes]** Appendix C specifies the split procedure, rollout-safe protocol, LightGBM settings, feature list, every conformal variant, all seven baselines, every metric, and the pinned data revision and software environment.
5. **Open access to data and code. [No]** The data is fully open (public BAGEN rollouts at a pinned revision). The code is not attached to this submission; the full pipeline will be released under an MIT licence on acceptance, together with a digest-pinned Dockerfile and a single entry point that regenerates and verifies every reported number. Appendix C specifies everything needed for an independent reimplementation in the meantime.
6. **Experimental setting/details. [Yes]** Section 4 and Table 1 give the environments and budgets; Appendix C gives the rest, including the exact sweep grids for hard-cap baselines, the feasibility gate, and δ.
7. **Experiment statistical significance. [Yes]** Every reported number is a mean over five rollout-level split seeds. Figure 2 carries across-seed standard-deviation error bars; Table 5 reports the same quantity as a $\pm$ beside each coverage value. Appendix C specifies the cluster bootstrap ($B = 500$, resampling whole rollouts).
8. **Experiments compute resources. [Yes]** Appendix C states that the entire pipeline is CPU-only: no GPU, no model weights, and no LLM training or querying. All experiments fit gradient-boosted trees on tables of at most a few thousand rows inside one digest-pinned container.
9. **Code of ethics. [Yes]** We have reviewed the NeurIPS Code of Ethics and believe this work conforms. The research uses only an existing, publicly released dataset of logged agent rollouts; it involves no human subjects and no new model release.
10. **Broader impacts. [Yes]** Section 6 contains an explicit broader-impact paragraph on wasted compute versus false-abort risk, with δ calibration as the mitigation.
11. **Safeguards. [N/A]** We release no model weights, no pretrained language model, and no scraped data. The artifact is CPU-only calibration code whose only effect is to stop an agent's own run earlier.
12. **Licenses for existing assets. [Yes]** BAGEN [13] is cited and pinned to an exact revision. Appendix C lists the software stack with exact versions and licences (LightGBM MIT; numpy, pandas, scipy, scikit-learn BSD-3-Clause).

13. **New assets. [N/A]** No new asset is released with this submission. The implementation will be released on acceptance; no new dataset is created.
14. **Crowdsourcing and research with human subjects. [N/A]** All experiments are offline analyses of pre-existing logged agent rollouts.
15. **IRB approvals. [N/A]** No human subjects were involved.
16. **Declaration of LLM usage. [N/A]** No LLM is a component of the core method. We neither train, fine-tune nor query any LLM in this work.